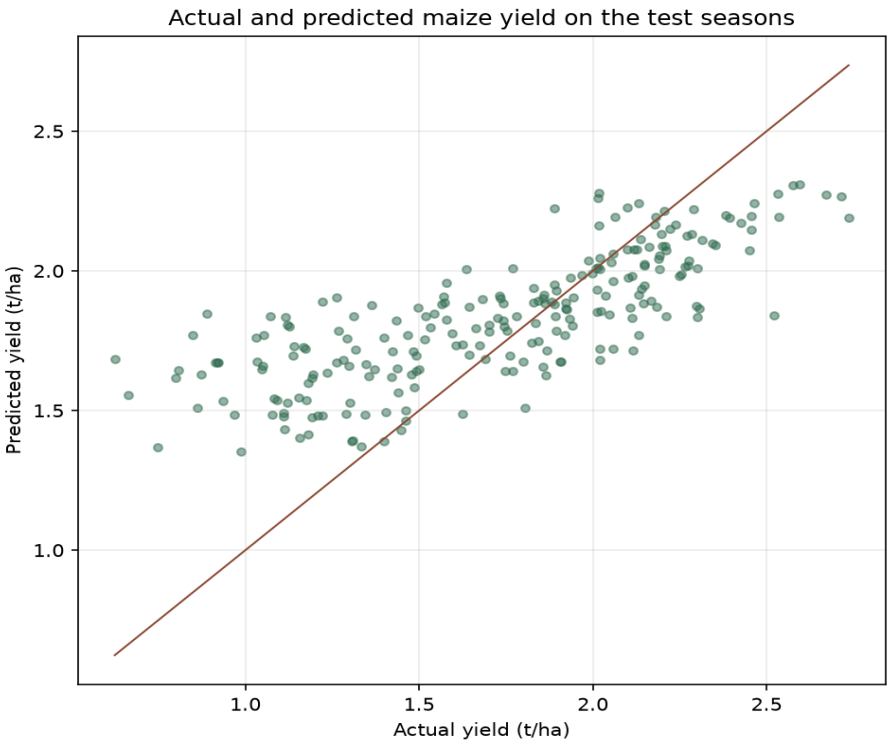
SmartCrop — maize yield model report

Macheke, Mashonaland East, Zimbabwe. Crop: maize. Target: yield_t_ha.

SYNTHETIC DATA. These rows were generated by the SmartCrop synthetic generator for software and model-development exercises. They are not measurements from Macheke, Mashonaland East, Zimbabwe, and they must not be cited as agricultural statistics.

Algorithm: Random Forest Regressor 1.0.0. Test MAE 0.264 t/ha, RMSE 0.342 t/ha, R^2 0.468. The academic 80% target is defined here as test-set R^2 of at least 0.80. R^2 is the coefficient of determination, not classification accuracy.



Model	Validation R^2	Validation RMSE	Validation MAE
Random Forest Regressor	-0.310	0.352	0.285
Gradient Boosting Regressor	-0.359	0.358	0.293
Decision Tree Regressor	-0.455	0.371	0.298
Linear Regression	-1.150	0.451	0.386

Top inputs by permutation importance. This is contribution to predictions, not causation.
1. fertilizer_kg_ha: 0.488
2. rainfall_mm: 0.156
3. planting_day: 0.064
4. soil_nitrogen: -0.001
5. avg_temperature_c: -0.003